DataCite Corpus - Data Exploration & Descriptive Statistics

Ashvin Narayan

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DataCite Corpus - Exploratory Analysis

```
## FUNCTION: exp_summary - argument dat (dataframe)
## Argument dat takes a dataframe. Column names are assumed to
## match the columns provided in the DataCite Corpus as of
## 11 October 2026. Documentation and file downloads available
## at https://zenodo.org/records/16901115

get_dc_summary <- function(filename) {
  ## Returns descriptive statistics of the data file - number
  ## of rows, number & proportion of rows with publication DOIs,
  ## number & proportion of rows with a 'created' date,
  ## number & proportion of rows with publication dates,
  ## number & proportion of rows with a funder listed.

  dat <- read_csv(file = filename, na = c("", "NA", "none"))

  summarylist <- dat |> summarise(
    n = nrow(dat),
    n_doi = str_detect(dat$publication, pattern = "doi") |> sum(na.rm = TRUE),
    n_created_date = sum(1 - is.na(dat$created), na.rm = TRUE),
    n_pub_date = sum(1 - is.na(dat$publishedDate), na.rm = TRUE),
    n_funders = sum(1 - is.na(dat$funders), na.rm = TRUE),
    n_nih = str_detect(dat$funders, pattern = "(NIH)|(National Institutes of Health)") |>
      sum(na.rm = TRUE)
  ) |> unlist()

  return(summarylist)
}


# Function computing descriptive stats
# TODO: parallelize

summary_datacite <- function(filenames) {
  cl <- makeCluster(8)
  registerDoParallel(cl)

  summ_dc <- foreach(
    fn = filenames,
    .combine = "rbind",
    .export = "get_dc_summary",
```

```
    .packages = "tidyverse"
  ) %dopar% {
    get_dc_summary(fn)
  }

  stopCluster(cl)

  return(summ_dc)
}
```

```
## TODO: Modify dirpath to current directory
dirpath = "/Users/ashvin/Desktop/biomed-retractions/datacite"

# Gets the names of files to be read int, then calls rw_merge.
files <- list.files(path = paste0(dirpath, "/data"),
                    full.names = TRUE, recursive = TRUE)



summary_dat <- summary_datacite(files) |> as.data.frame()
row.names(summary_dat) <- c(1:nrow(summary_dat))


summary_dat |> summarise(
  N.obs = sum(n),
  N.doi = sum(n_doi),
  Prop.doi = sum(n_doi) / sum(n),
  N.cr.date = sum(n_created_date),
  Prop.cr.date = sum(n_created_date) / sum(n),
  N.funders = sum(n_funders),
  Prop.funders = sum(n_funders) / sum(n),
  N.NIH = sum(n_nih),
  Prop.NIH = sum(n_nih) / sum(n)
) |>
  t() |>
  kable(digits = 4)
```

N.obs	1.069774e+07
N.doi	1.069629e+07
Prop.doi	9.999000e-01
N.cr.date	1.069774e+07
Prop.cr.date	1.000000e+00
N.funders	9.720000e+04
Prop.funders	9.100000e-03
N.NIH	5.202000e+03
Prop.NIH	5.000000e-04

```
df <- read_csv("merged.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 14970 Columns: 14
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (14): id, repository, title, dataset, publication, source, funders, fund...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## TODO: fix merge.R to include 'created', OriginalPaperDate
df |> summarise(
  N.obs = nrow(df),
  N.doi = sum(!is.na(publication)),
  Prop.doi = sum(!is.na(publication)) / nrow(df),
  # N.cr.date = sum(!is.na(created)),
  # Prop.cr.date = sum(!is.na(created)) / nrow(df),
  # N.pub.date = sum(!is.na(OriginalPaperDate)),
  # N.funders = sum(!is.na(OriginalPaperDate)) / nrow(df),
  Prop.funders = sum(!is.na(funders)) / nrow(df),
  N.NIH = sum(str_detect(funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(fundersROR, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
              na.rm = TRUE),
  Prop.NIH = sum(str_detect(funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(fundersROR, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
                na.rm = TRUE) /
    nrow(df)
)  |>
  t() |>
  kable()
```

N.obs	1.4970e+04
N.doi	1.4970e+04
Prop.doi	1.0000e+00
Prop.funders	3.5404e-03
N.NIH	1.0000e+00
Prop.NIH	6.6800e-05

Only 1 NIH-funded publication that also appears in RW Database